A Physics-Aware Benchmark for Phase-Resolved Neural Operator Learning in Boiling Flows: Statistically Rigorous Evidence for Condition-Dependent Architecture Trade-offs

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Abstract

Neural surrogates for boiling flows must forecast smooth thermal structure, sharp liquid-vapor interfaces, and locally conservative transport over long autoregressive horizons. We introduce a reproducible, statistically rigorous benchmark for this task and use it to show that architecture rankings for phase-resolved boiling forecasting are highly condition-dependent — a finding with direct implications for how the field should evaluate operator-learning claims going forward. On a fixed tutorial-scale trajectory split, a Tucker-factorized Fourier Neural Operator (T-FNO) and a parameter-comparable U-Net exhibit a statistically significant Pareto trade-off (paired bootstrap intervals, exact sign-flip tests, Holm-Bonferroni correction, n=11 seeds): T-FNO is significantly better on interface-temperature fidelity, U-Net is significantly better on mass-conservation error. When this same comparison is repeated on two independently held-out physical conditions from the same source dataset, however, the trade-off does not reproduce: U-Net is descriptively favored over T-FNO on every primary metric, including interface temperature, though the five-seed independent test is statistically underpowered. A predeclared spectral divergence-penalty loss, applied to a local-global hybrid architecture, restores mass-conservation non-inferiority to U-Net on the original split (p=.000488) and transfers the conservation advantage directionally to the

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independent conditions, though not yet with confirmatory statistical power there. We additionally identify and correct a previously unreported evaluation pitfall — phase-indicator (vapor-fraction) predictions exceeding their physical [0,1] range during autoregressive rollout, which we show can materially confound downstream interface and dry-area diagnostics — and demonstrate a predeclared statistical decision-gate methodology that correctly rejects a promising but non-replicating single-seed observation. A critical-heat-flux (CHF)-motivated heater-adjacent dry-area proxy is evaluated across all trajectories but reveals no sustained ground-truth event in any of them, so no claim of validated CHF detection is made. We contribute the benchmark protocol, the physical-validity check, the decision-gate methodology, and — as the central empirical finding — direct evidence that single-condition architecture comparisons in this domain risk reporting a property of the test trajectory rather than a genuine architectural property, with a concrete demonstration of how to detect this via independent-condition replication.

Keywords: neural operator, Fourier neural operator, pool boiling, phase-resolved forecasting, conservation laws, BubbleML

S. B. Mahafuj Bondhon, Department of Mechanical Engineering, Bangladesh University of Engineering and Technology (BUET), Ramna, Dhaka-1000, Bangladesh. Corresponding author: `2210062@me.buet.ac.bd`. *ORCID: 0009-0009-6695-365X.* *The bibliography entries in the generated manuscript were checked against the primary records documented in* `BIBLIOGRAPHY_VERIFICATION.md`. *The prior public repository release is* `https://github.com/RealmeBTI/bubbleml-submission/releases/tag/v1.0.4`*; its matching Zenodo version archive is* `https://doi.org/10.5281/zenodo.21858198`. *The archive/version scope and the retained-artifact limitations are recorded in* `PRESUBMISSION_FINAL_CHECK.md`.

1. Introduction

Pool boiling is central to high-power electronics cooling, data-center thermal management, and nuclear thermal-hydraulics, where heat fluxes exceed single-phase capabilities. The mechanism's effectiveness is bounded by the critical heat flux (CHF)—the point at which a growing vapor film insulates the heated surface, triggering runaway temperature excursions. Classical correlations (e.g., Zuber's hydrodynamic model) estimate aggregate CHF thresholds but do not resolve transient local bubble dynamics. While deployment-grade CHF forecasting requires synchronized wall-heat-flux measurements and verified boiling-crisis transitions—neither of which is present in available open simulation benchmarks—resolving localized, heater-adjacent dry-area dynamics serves as a diagnostic for phase-interface integrity. In this work, we evaluate a heater-adjacent dry-area fraction proxy across stable boiling trajectories strictly as an autoregressive physical-stability test for neural operator rollouts, explicitly distinguishing phase-validity stress-testing from validated CHF event detection.

High-fidelity interface-tracking computational fluid dynamics (CFD) can resolve bubble nucleation, growth, coalescence, and departure, but its cost limits repeated design exploration and real-time digital-twin use. Operator learning offers a route to fast state-to-state surrogates learned directly from CFD trajectories. The Fourier Neural Operator (FNO) is an attractive candidate because it represents global spatial coupling through a resolution-invariant spectral representation. Boiling fields, however, combine non-periodic domain boundaries with a sharp, near-discontinuous liquid-vapor interface — properties in tension with FNO's implicit periodicity assumption and its low-frequency spectral truncation. The BubbleML benchmark reported that a vanilla FNO underperforms a U-Net on pool-boiling temperature forecasting by roughly a factor of two in boundary RMSE (not the order-of-magnitude gap sometimes assumed in early framing of this problem), attributed tentatively to convolutional edge sensitivity versus the Fourier layer's periodicity assumption.

This motivates a natural question: do parameter-efficient FNO variants close this gap on the full, phase-resolved multi-field boiling state (not just temperature), and — critically — **is any observed architecture ranking a genuine, generalizable property of the architectures, or an artifact of the specific trajectory it was measured on?** The second half of this question is, to our knowledge, not systematically addressed in prior boiling-operator benchmark work, which typically reports single-trajectory or single-seed comparisons. We treat it as a first-class experimental question rather than an assumed non-issue.

This paper makes five contributions:

1. A reproducible, physics-aware benchmark protocol for five-field (velocity, pressure gradient, temperature, vapor-phase indicator) phase-resolved boiling forecasting, evaluated with paired bootstrap intervals, exact sign-flip tests, and Holm-Bonferroni correction across multiple seeds.
2. Identification and correction of a previously unreported evaluation pitfall: phase-indicator predictions exceeding their physical [0,1] range under autoregressive rollout, and a demonstration that this materially confounds downstream evaluation metrics that depend on the phase field.
3. A predeclared statistical decision-gate methodology, demonstrated concretely by its correct rejection of a promising but non-replicating single-seed rollout observation — a discipline we argue should be standard practice before committing engineering effort to an observed effect.
4. **A cross-condition replication check**, in which a statistically confirmed architecture trade-off (T-FNO favors interface fidelity, U-Net favors mass conservation) on one trajectory split is re-tested under an independent protocol on two held-out conditions from the same source dataset — and does not reproduce. The protocol also differs in spatial resolution, so this is evidence against generalizing the tutorial ranking, not evidence that wall temperature alone caused the ranking change.
5. A predeclared, mechanistically motivated conservation intervention (a spectral velocity-divergence penalty) that restores mass-conservation non-inferiority to U-Net on the original split and transfers the conservation advantage directionally — though not yet confirmatorily — to the independent conditions, illustrating how a targeted physical constraint, rather than architecture complexity alone, is needed to move a specific objective.

Consistent with contribution 4, our central finding is not a single winning architecture. The tutorial ranking did not replicate under an independent cross-condition protocol that also differed in spatial resolution. Reporting a ranking from a single trajectory—however statistically rigorous the seed-level inference—is therefore not sufficient evidence of a generalizable architectural property. This is a methodological finding of independent value to the broader neural-operator-for-science community, beyond its specific application to boiling.

2. Related Work

2.1 Classical CHF prediction and boiling simulation

Critical heat flux has been studied mechanistically since the mid-twentieth century, most influentially through Zuber's hydrodynamic instability model[1]. Engineering practice also uses application-specific correlations for CHF estimation. These are computationally simple aggregate predictors, but they do not resolve transient local bubble dynamics on a specific surface and offer no natural mechanism for real-time, sensor-driven early warning.

Direct numerical resolution of boiling requires interface-tracking methods — volume-of-fluid, level-set, or phase-field formulations coupled to the Navier-Stokes and energy equations. The BubbleML dataset[2], which underlies the present work's data, was generated with the Flash-X interface-tracking solver and spans multiple fluids, heater geometries, and boiling regimes. Its accompanying benchmark paper is, to our knowledge, the first systematic comparison of neural operator architectures against convolutional baselines on boiling data, and its reported ~2x FNO-versus-U-Net temperature BRMSE gap is the empirical starting point for the present investigation.

2.2 Operator learning and convolutional surrogates

The Fourier Neural Operator[3] performs global convolution via pointwise multiplication in a truncated Fourier domain, and has demonstrated strong performance on smooth, largely periodic PDE systems. Convolutional U-Net architectures[4] instead rely on local receptive fields and multi-scale skip connections, and serve as a strong general-purpose baseline for dense spatial prediction, including PDE surrogate modeling.

Several parameter-efficient FNO variants address the capacity and generalization trade-offs of the original architecture. Tensor-factorized (Tucker) FNO represents the spectral weight tensor in a low-rank Tucker decomposition; axis-factorized FNO (F-FNO)[5] factorizes the spectral convolution along each spatial axis independently. The BubbleML benchmark's own results indicate both variants outperform vanilla FNO on boiling forecasting despite having substantially fewer parameters, attributed to reduced overfitting to excess spectral capacity rather than a fix to the underlying periodicity or discontinuity issue. U-shaped FNO (UNO)[6] grafts U-Net-style multi-scale structure onto Fourier blocks; the LOGLO-FNO family of Kalimuthu et al. (2025)[7] explicitly combines local and global spectral branches to improve local, high-frequency representation. Transformer-based operators, including a recent boiling-specific transformer forecaster (Bubbleformer[8]) benchmarked on the extended BubbleML 2.0 release, represent a further active direction — indicating this remains a contested design space rather than a settled one. None of these architecture families, to our knowledge, has been evaluated with an explicit cross-condition replication check of the kind this paper performs. Hard-constrained alternatives to the present soft divergence penalty include the continuity-by-construction networks of Richter-Powell et al. (2022)[9] and the differentiable spectral Leray projection used by Li et al. (2026)[10]; neither is evaluated here on phase-resolved boiling.

2.3 Experimental boiling diagnostics

Independent of the simulation-data operator-learning literature, Ravichandran et al. (2021)[11] used high-resolution infrared thermometry to investigate boiling-crisis diagnostics, and Ravichandran et al. (2023)[12] developed online dry-area detection from infrared measurements using deep learning. This experimental line of work and the simulation-based operator-learning line of work reviewed above remain largely disjoint; no result in the present paper establishes transfer to experimental measurements or deployment-grade CHF warning.

2.4 Reproducibility and the gap addressed

Open boiling datasets are fragmented across repositories, fields, grids, regimes, and evaluation conventions, a fragmentation documented by Dunlap et al. (2026)[13] in their survey of open multimodal thermal-fluid datasets and software. This work's explicit, checksum-verified data and evaluation pipeline is designed to partially mitigate this within its own declared artifact boundary. Combined with the cross-condition replication check described above, we position this work primarily as a benchmark and evaluation-methodology contribution — a genuinely rigorous protocol for making and testing architecture claims on chaotic, multiphysics surrogate tasks — rather than as a claim of a novel architecture. The individual architectural components used here (Tucker/factorized FNO, local-global hybridization, spectral divergence penalties, bounded output heads) are each drawn from or closely related to existing techniques; the contribution is their careful, statistically disciplined integration and, specifically, the demonstration that even a well-powered single-condition comparison among them is not sufficient evidence of a generalizable ranking.

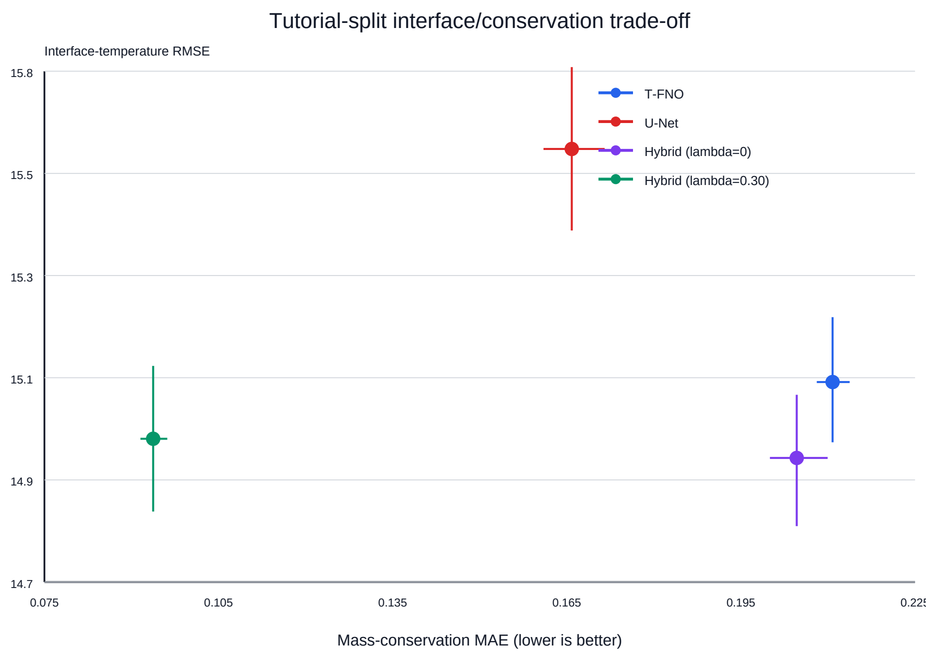

Figure 1: The retained tutorial-split results place T-FNO and U-Net on opposing sides of the interface-fidelity/mass-conservation trade-off; the local-global variants are shown for the intervention context. This figure summarizes the fixed-split result only and does not establish a cross-condition ranking.

Feature	BubbleML (original)	This work
Multiple trajectories	Yes (79 simulated conditions)	Yes
Interface and conservation metrics jointly reported	Not reported for the neural-PDE benchmark	Yes
Multi-seed paired statistical testing	Not reported	Yes (n=11 tutorial; n=5 cross-condition)
Cross-condition architecture replication	Not reported; the paper reports heat-flux holdout cross-validation	Yes
Physical output-validity enforcement	Not reported	Yes (bounded-alpha head)
Predeclared decision gates	Not reported	Yes
Artifact release	Partial (data, code, and model zoo reported)	Partial (stored-result self-test; raw data/checkpoints remain external)

Table 1. Protocol comparison. "Not reported" indicates that the original BubbleML paper does not document the feature for its neural-PDE benchmark; it does not establish absence.

3. Methods

3.1 Data, preprocessing, and splits

Tutorial-scale split. The primary comparison uses three official downsampled Pool-Boiling Subcooled FC-72 examples at 48×48 resolution, preprocessed into PyTorch tensors with no synthetic fallback. The trajectory-level split prevents temporal leakage: Twall-103 is train, Twall-106 is validation, and Twall-100 is test. Each source has 169 released-grid frames, yielding 160 valid five-history/five-future temporal windows per split.

Source	SHA-256	Role
`Twall-103.hdf5`	`25b305661dee59cf5df49eb2563a4ed08f 79664ba377e57877fcda5a948956dd`	train
`Twall-106.hdf5`	`4308e5be884c3edcf301e9c12bff8d499d 53729be3820aa18e255a5152fbd004`	validation
`Twall-100.hdf5`	`5bf2539628c3595f39517c466f6971da76 f137358771592c14ad1a5881e4d1bf`	test / rollout

The released fields are time×y×x grids; row 0 is immediately above the heater; physical boundary cells are omitted. Reported outer-grid quantities are therefore interior-edge proxies, not wall residuals.

Independent multi-trajectory split. The cross-condition replication check uses the official ten-trajectory, native 384×384 Pool-Boiling Subcooled legacy archive (SHA-256 `2eba140d74cbb7b01a55f0684227dea94306aea51 6a0f864831485d31d25f655`; 201 frames/trajectory; Twall 79, 81, 85, 90, 95, 98, 100, 103, 106, 110). Trajectory roles were frozen before any model training or test inspection: train on Twall 79, 85, 90, 95; validate on Twall 81; test independently on Twall 98 and Twall 110. Twall 100, 103, and 106 were excluded, since they form the tutorial split. Continuous fields were bilinearly downsampled to 96×96; the binary phase mask used nearest-neighbor downsampling; physical grid spacings were rescaled consistently. After discarding source frames 0–29, each trajectory contributes 170 frame records, yielding 644/161/322 valid windows for train/validation/test. The distinct tutorial and cross-condition protocols are visualized in Figure 2. A separate one-epoch, batch-size-one feasibility check was also run at native 384×384 resolution to test direct execution and memory feasibility only (not convergence or ranking).

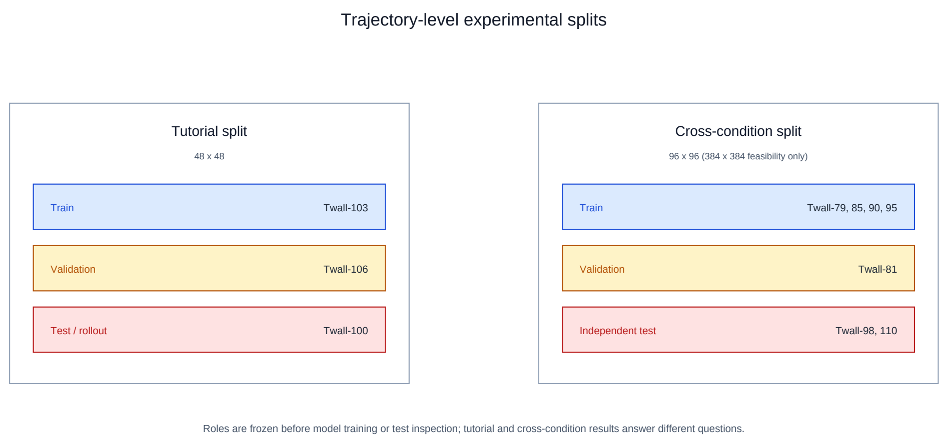

Figure 2: The tutorial split evaluates a fixed three-trajectory comparison; the independent cross-condition protocol tests whether that comparison persists on two held-out conditions. The protocols also differ in spatial resolution (48×48 versus 96×96), so the comparison does not isolate a wall-temperature effect. Roles were frozen before training or test inspection.

3.2 Model architectures and training protocol

Four base models are compared: vanilla FNO, Tucker-factorized FNO (T-FNO, rank 0.1), axis-factorized FNO (F-FNO)[5], and a parameter-comparable U-Net. Each uses five history and five forecast frames (25 input/output channels), AdamW (lr 1e-3, weight decay 0.01), batch size 8, horizontal reflection augmentation, global gradient-norm clipping at 1.0, and training to a predeclared two-window validation-plateau stopping rule (200-epoch ceiling). Fourier-family models train at full 48×48 (tutorial) or 96×96 (cross-condition) resolution with 24×24 real-FFT modes — the maximum representable given the Nyquist limit at these resolutions (max unique modes = N/2 + 1), correcting an earlier invalid 64-mode configuration that would have silently collapsed under half-resolution training.

The tutorial-split four-model comparison used seeds 42, 100, 1234, 2025, 9999; a disclosed six-seed T-FNO/U-Net extension added seeds 7, 17, 314, 2718, 4242, 7777, for 11 paired seeds total. The cross-condition comparison used seeds 42, 100, 1234, 2025, 9999 (n=5) on an actual NVIDIA Tesla T4 via CUDA; tutorial-split runs used an Apple M2 via Metal Performance Shaders (MPS). With five paired seeds, the minimum attainable two-sided exact sign-flip p-value is 0.0625; direction and interval width, not corrected significance, are the primary evidence at this sample size.

3.3 Physical output-bounding for the phase indicator

Unconstrained models produced decoded vapor-fraction (alpha) predictions outside the physical [0,1] interval during autoregressive rollout (observed ranges up to [-0.157, 1.408]). We correct this at the model level: writing the raw normalized-space alpha logit as z_α, and a_0, a_1 as the normalized coordinates that decode to physical alpha 0 and 1 respectively, the bounded output is

$$\widehat{\alpha}_N = a_0 + (a_1 - a_0)\,\sigma(z_\alpha), \qquad \widehat{\alpha} = \mathrm{decode}(\widehat{\alpha}_N) \in [0, 1]. \tag{1}$$

All non-alpha channels remain unconstrained. This guarantees physical validity by construction, without post-hoc clipping of autoregressively fed-back predictions.

3.4 Local-global hybrid architecture

For hybrid layer ℓ, the local-global composition adds a learned local convolution branch in parallel with the spectral and pointwise-residual paths:

$$z^{(\ell+1)} = \phi\Big(\mathcal{K}^{(\ell)}_{\mathrm{Tucker}} z^{(\ell)} + \mathcal{C}^{(\ell)}_{3\times3} z^{(\ell)} + W^{(\ell)} z^{(\ell)}\Big)\,, \tag{2}$$

where $\mathcal{K}_{\mathrm{Tucker}}$ is the truncated Tucker-factorized spectral convolution, $\mathcal{C}_{3\times3}$ is a learned 3×3 local convolution, W is the pointwise residual map, and ϕ is the layer nonlinearity. This architecture was motivated specifically by the confirmed tutorial-split Phase 1 Pareto trade-off (Section 4.1) — a predeclared statistical decision gate (Section 3.6) had separately and correctly rejected an earlier, unrelated intervention aimed at a non-replicating single-seed rollout artifact; the two are distinct and should not be conflated.

3.5 Conservation-targeted divergence penalty

The conservation intervention augments the local-global hybrid's training objective with a spectral velocity-divergence penalty, computed on decoded physical-coordinate velocity to avoid spatial discretization error:

$$\mathcal{L} = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{div}} \cdot \frac{1}{BTHW} \sum_{b,t,x,y} \left| \mathcal{F}^{-1}\big[ik_x \mathcal{F}(u_{b,t}) + ik_y \mathcal{F}(v_{b,t})\big]_{x,y} \right|, \tag{3}$$

using each sample's physical grid spacings for the wavenumbers k_x, k_y. Validation data MSE remains the checkpoint-selection criterion; λ_{div} was selected in two stages, described fully in Section 4.3, to avoid conflating hyperparameter selection with confirmatory evaluation.

3.6 Statistical protocol and decision gates

All architecture and intervention claims use paired, multi-seed inference: a deterministic 10,000-sample paired bootstrap for confidence intervals, an exact paired sign-flip test for significance, and Holm-Bonferroni correction across the full family of metrics tested in a given comparison. An earlier statistical audit corrected two implementation details — removing an unnecessary Monte Carlo add-one correction from the exact sign-flip enumeration, and excluding compute-only (non-error) metrics from the Holm family — and all numbers reported in Section 4 reflect the corrected implementation.

Architectural interventions are evaluated under **predeclared decision gates**: a fixed statistical criterion (non-inferiority margin, significance threshold, and metric family) is frozen before training and evaluation, and the intervention is only reported as successful if the frozen criterion is met. This discipline was demonstrated concretely when an initial single-seed pair showed a promising reversal in rollout dry-area tracking after the phase-bounding fix (Section 3.3); a predeclared 11-seed confirmatory test showed this reversal did not replicate (Section 4.2), and the corresponding architectural interventions were correctly not pursued on that basis. The local-global hybrid (Section 3.4) and divergence penalty (Section 3.5) are motivated by a separate, independently confirmed finding (the tutorial-split Pareto trade-off) and are not affected by that earlier gate's rejection.

3.7 CHF-motivated autoregressive proxy evaluation

A fully autoregressive evaluator feeds each model's own five-frame predictions back as input for the next step (no ground-truth injection), producing long rollouts from a short seed history. The CHF-motivated proxy signal is the fraction of heater-adjacent cells (rows 0:4) with predicted alpha > 0.5. The event threshold is $\max(0.10, \text{median of first 20 ground-truth forecast frames} + 0.10)$, requiring three consecutive frames at or above threshold to count as a sustained precursor. This is an illustrative phase-based proxy, not a calibrated CHF label: none of the available trajectories has a synchronized wall-heat-flux series or an independently verified stable-to-CHF transition. Figure 3 locates this evaluation within the retained-artifact workflow.

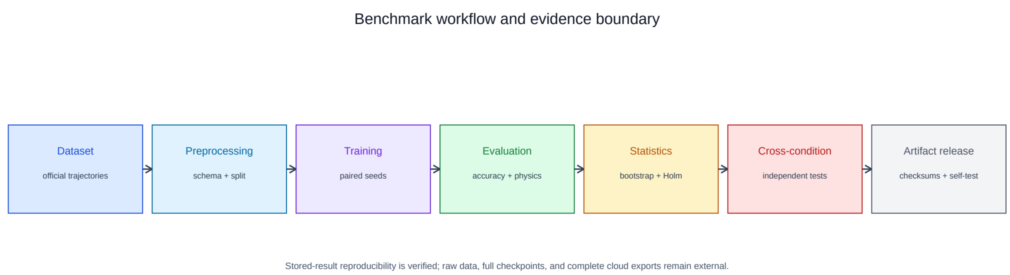

Figure 3: Benchmark workflow. The last transition makes the artifact boundary explicit: the stored-result statistical self-test is reviewer-runnable, while raw data, full checkpoints, and complete cloud-side cross-condition exports remain external.

3.8 Exploratory same-trajectory resolution comparison

As a post-hoc exploratory observation, T-FNO and U-Net were also trained and evaluated on the same three tutorial trajectories (Twall-103/106/100) at 96×96 resolution using identical random seeds (n=11 paired seeds: 7, 17, 42, 100, 314, 1234, 2025, 2718, 4242, 7777, 9999) and the same model configurations and evaluation metrics as the 48×48 tutorial split. This configuration is archived on the `experiment/resolution-control-n11` branch.

This comparison is **not** a controlled resolution experiment. Multiple factors differ between the 48×48 and 96×96 configurations beyond spatial resolution: the temporal frame ranges differ (169 frames, timesteps 30–198, at 48×48 versus 166 frames, timesteps 30–195, at 96×96), the hardware environments differ (Apple M2/MPS at 48×48 versus NVIDIA Tesla T4/CUDA at 96×96), source code revisions differ, and 48×48 model checkpoints are retained whereas the 96×96 checkpoints are not available in the authorized artifact set. The result should be interpreted as a descriptive observation across two stored pipeline configurations that differ in resolution among other factors, not as evidence that spatial resolution alone caused any observed difference. A full control-matrix documenting all factor differences is retained in `audit_n11/control_matrix.md`.

4. Results

4.1 Tutorial split: T-FNO and U-Net form a statistically confirmed Pareto trade-off

Across 11 paired seeds, no overall GWRMSE winner was established (T-FNO - U-Net: -0.05698, 95% paired bootstrap CI [-0.12314, +0.00946], sign-flip p=.1469, Holm p=1.0). T-FNO was significantly better on interface-temperature RMSE (-0.50190, Holm p=.0352) and interface-temperature-jump MAE (-0.43147, Holm p=.0215), but significantly worse on mass-conservation MAE (+0.04494, Holm p=.0215). The fixed-split trade-off is shown in Figure 1.

4.2 Physical validity is an evaluation requirement, independent of architecture ranking

Unbounded models produced physically invalid alpha values under rollout ([-0.157, 1.408] for T-FNO, [-0.108, 1.417] for U-Net over 164 steps). The output-head fix (Eq. 1) constrained both to [0,1] by construction. An initial single-seed comparison after this fix showed a large reversal in cumulative dry-area tracking error (bounded T-FNO 0.12167 vs. bounded U-Net 0.05275); a predeclared 11-seed confirmation, however, found this did not replicate as a significant effect (paired difference +0.00775, 95% CI [-0.00393, +0.01829], Holm p=1.0 across all 21 tested rollout metrics, smallest adjusted p=0.3997). We report this explicitly as a methodological finding: physical-validity correction and architecture superiority are distinct questions, and single-seed rollout observations — even after a legitimate bug fix — require the same multi-seed confirmatory discipline as any other architecture claim.

4.3 A local branch alone does not recover conservation; a targeted divergence penalty does, on the tutorial split

The zero-penalty local-global hybrid (Eq. 2) was tested under three predeclared non-inferiority criteria (5% margins from the Phase 1 comparators): non-inferior to T-FNO on interface-temperature RMSE (Holm p=.00146) and jump MAE (Holm p=.00146, bootstrap interval favoring the hybrid), but it **failed** mass-conservation non-inferiority to U-Net by a wide margin (+0.03876 against a 0.00829 margin; Holm p=1.0 in the unchanged complete-metric family, p=.02148). A local receptive field alone does not import U-Net's conservation behavior.

An initial divergence-penalty pilot (2 seeds, $\lambda_{\text{div}} \in \{.01, .03, .10\}$, all eligible under a 5% data-fit guard) selected $\lambda_{\text{div}} = .10$; an 11-seed confirmation passed the single predeclared mass-conservation non-inferiority criterion (mean mass MAE 0.14499 vs. U-Net's 0.16586; paired difference -0.02086, 95% CI [-0.02845, -0.01331]; p=.000488), without an established interface regression relative to the zero-penalty hybrid.

An expanded, more rigorous sensitivity sweep (3 seeds, $\lambda_{\text{div}} \in \{.01, .03, .10, .20, .30\}$, both a data-fit and an interface-RMSE guard) found validation divergence decreasing monotonically through the full tested range, with 0.30 — the upper tested boundary — selected as the eligible candidate with lowest divergence. Figure 4 shows the retained sweep: the selected value lies at the tested upper boundary. **This selection identifies a value that works within the tested range; it does not identify an interior optimum, and behavior beyond 0.30 remains untested** (see Section 6, future work). A fresh 11-seed confirmation at $\lambda_{\text{div}} = .30$ passed the same predeclared criterion with a substantially larger margin: mean mass MAE 0.09373 vs. U-Net's 0.16586 (paired difference -0.07212, 95% CI [-0.07940, -0.06535], p=.000488), with no established interface regression relative to the zero-penalty hybrid. We adopt $\lambda_{\text{div}} = .30$ as the primary divergence-hybrid configuration for the remainder of this paper.

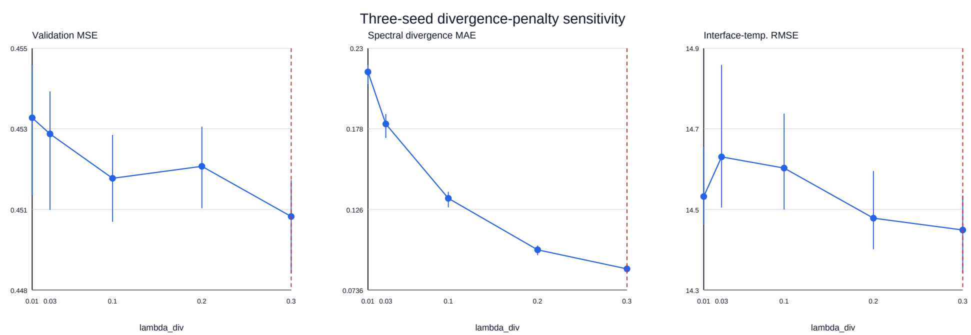

Figure 4: The expanded sensitivity sweep reports validation MSE, spectral divergence MAE, and interface-temperature RMSE for each tested divergence-penalty weight. The red line marks the selected upper-bound value, $\lambda_{\text{div}} = .30$; the sweep does not establish behavior beyond that range.

4.4 Cross-condition replication: the tutorial-split trade-off does not reproduce (central finding)

On two independently held-out physical conditions (Twall 98, Twall 110; n=5 paired seeds), the descriptive ranking changed substantially from the tutorial split:

Model	GWRMSE	Interface-temperature RMSE	Interface-temperature-jump MAE	Mass-conservation MAE
T-FNO	11.5974	14.5207	8.7655	0.12989
U-Net	**11.5029**	**13.9057**	**8.5757**	0.11560
Local-global hybrid (zero penalty)	11.6323	14.4567	8.7021	0.13583
Divergence hybrid ($\lambda_{\text{div}} = .30$)	11.5859	14.2120	8.6981	**0.08669**

For T-FNO minus U-Net: GWRMSE +0.09448 (95% CI [+0.04363, +0.14534]), interface-temperature RMSE +0.61497 ([+0.31906, +0.90941]), jump MAE +0.18985 ([+0.18080, +0.19935]), mass MAE +0.01429 ([+0.00915, +0.02085]). Every unadjusted exact two-sided p-value is .0625 (the minimum attainable at n=5) and every Holm-adjusted p-value is 1.0. **The directions are consistent across all seeds and both held-out trajectories, and they consistently favor U-Net — the opposite of the tutorial split's interface-fidelity result** — but the five-seed design cannot establish corrected statistical significance. We report this as a clear descriptive reversal that current statistical power cannot confirm, not as an established new ranking.

The divergence hybrid's conservation advantage transferred directionally: mass MAE 0.08669 vs. U-Net's 0.11560 (paired difference -0.02891, 95% CI [-0.03359, -0.02422] relative to U-Net — descriptive, as this specific superiority contrast was not the split's predeclared target), while remaining descriptively worse than U-Net on GWRMSE, interface-temperature RMSE, and jump MAE. The per-trajectory breakdown (Twall 98 and Twall 110 separately) shows the same ordering within each condition, indicating the reversal is not driven by one anomalous trajectory, though two conditions remain too few to characterize population-level physical-regime variation.

4.5 CHF-motivated proxy: no sustained event in any available trajectory

The dry-area proxy (Section 3.7) was evaluated on Twall-100 (tutorial test) and, newly, on the independent Twall-98 and Twall-110 trajectories. None showed a sustained crossing: Twall-100 had 3/164 above-threshold frames (longest run 2), Twall-98 had 0/170, and Twall-110 had 1/170 (longest run 1). Figure 5 shows the retained rollout traces and thresholds. No trajectory in the currently available data supports a lead-time, sensitivity, or missed-event-rate evaluation. This is a negative feasibility result for CHF-proxy event evaluation with current data, not evidence about CHF predictability in general.

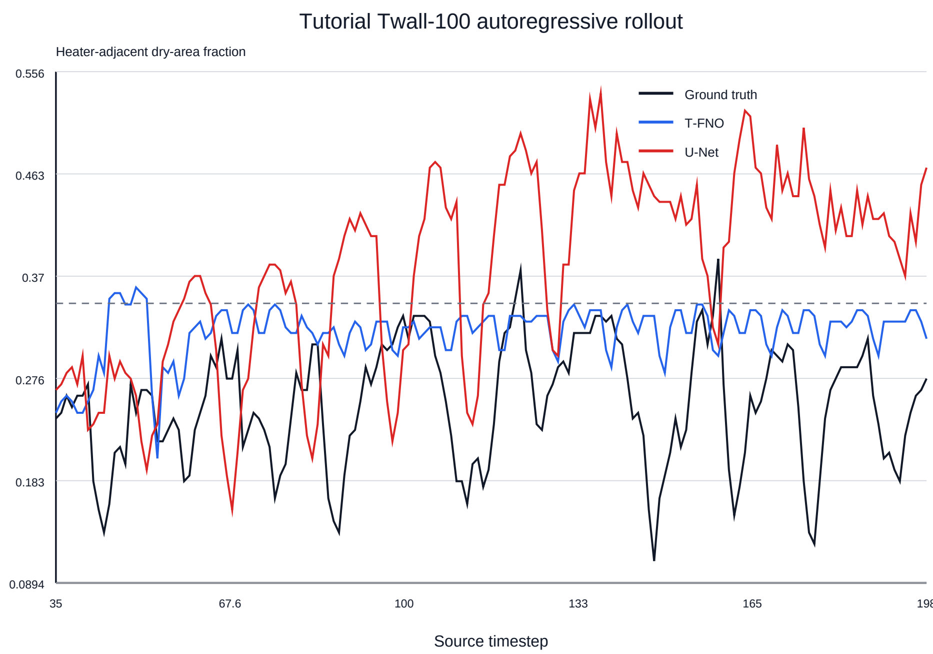

Figure 5: Heater-adjacent dry-area proxy traces for the available tutorial and held-out trajectories. The plotted thresholds are not crossed for three consecutive frames, so none of these trajectories supports a sustained-event evaluation.

4.6 Native-resolution feasibility and computational cost

One-epoch, batch-size-one runs at native 384×384 completed successfully for all four models on the Tesla T4 (5.91–6.99s/epoch for the Fourier-family models, 2.77s for U-Net), confirming direct execution feasibility without establishing convergence or ranking. At the converged 96×96 cross-condition scale:

Model	Parameters (stored / real-scalar)	Mean training time/seed (s)	Inference latency (ms/window)	Throughput (windows/s)
T-FNO	548,837 / 1,040,625	216.09	4.467	223.94
U-Net	7,770,169 / 7,770,169	98.72	2.026	493.63
Local-global hybrid	696,549 / 1,188,337	293.45	5.369	186.25
Divergence hybrid	696,549 / 1,188,337	294.43	5.384	185.75

U-Net is the fastest model to train and the lowest-latency at inference, despite having roughly 6–14× more parameters than the FNO-family variants — a practical consideration alongside the accuracy/conservation trade-offs above.

The retained tutorial validation histories are shown in Figure 6. They document the available tutorial training runs but do not establish cross-condition convergence, because the complete cross-condition per-seed histories are not retained locally.

4.7 Exploratory cross-resolution comparison on tutorial trajectories (not a controlled resolution experiment)

For the same tutorial trajectories (Twall-103/106/100) trained at 96×96 resolution (Section 3.8), the stored n=11 paired T-FNO minus U-Net differences are:

- GWRMSE: +0.10868 (Holm $p = 0.00391$; favors U-Net)
- Interface-temperature RMSE: +0.46219 (Holm $p = 0.00977$; favors U-Net)
- Interface-temperature-jump MAE: +0.21433 (Holm $p = 0.00391$; favors U-Net)

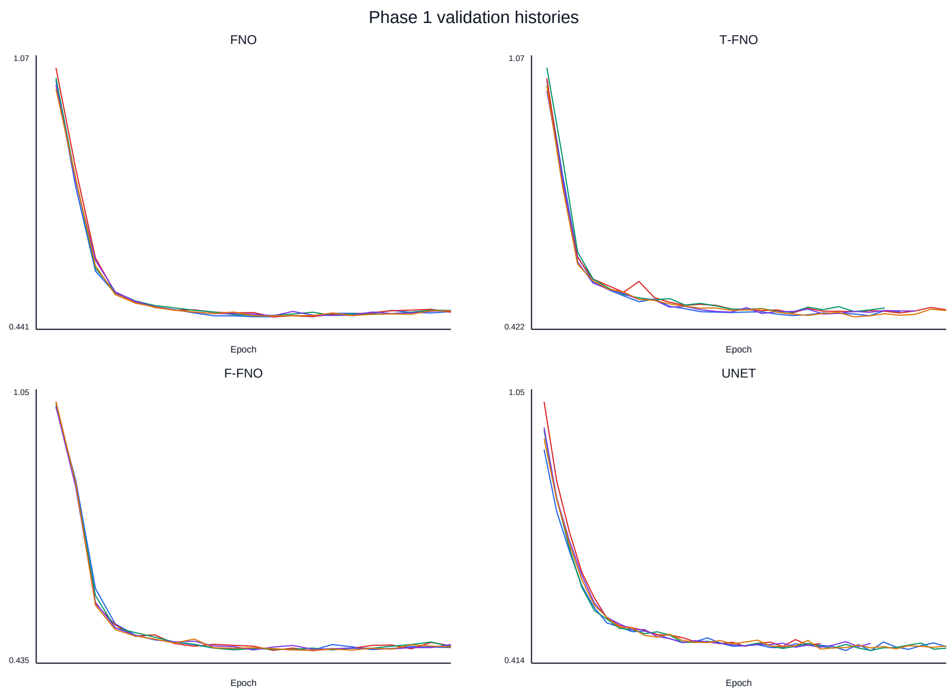

Figure 6: Per-seed validation histories for the retained Phase 1 tutorial-split runs. These curves provide training-trace context for the stored tutorial results only; they are not a substitute for unavailable cross-condition training histories.

- Mass-conservation MAE: +0.04906 (Holm $p = 0.00391$; favors U-Net)

(Paired bootstrap confidence intervals were not computed for this exploratory set, as the inference is entirely descriptive.)

All four metrics favor U-Net in the 96×96 stored configuration, with Holm-adjusted p-values well below 0.05 — a marked contrast to the 48×48 tutorial split, where T-FNO was significantly better on interface-temperature metrics (Section 4.1). For mass conservation, both resolutions return a positive difference (T-FNO worse than U-Net), so the direction holds across configurations for that metric; only the magnitude differs (+0.04494 at 48×48 versus +0.04906 at 96×96).

These differences are stored-pipeline observations that reflect the combined effect of resolution and the other pipeline factors documented in Section 3.8 and `audit_n11/control_matrix.md`. They cannot be attributed to spatial resolution as an isolated cause. We report them explicitly as exploratory because the statistical power (n=11) is sufficient to detect the signed direction reliably, but the absence of experimental controls prevents any causal interpretation. The correct scientific reading is: *the relative T-FNO/U-Net paired differences observed in the 96×96 stored pipeline configuration are consistently opposite in sign to the 48×48 interface-fidelity results, and this observation motivates a future dedicated study that would need to control the confounding pipeline factors before attribution is possible.*

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5. Discussion

5.1 Condition dependence is the central finding, not a limitation to route around

The tutorial-split result (Section 4.1) is statistically confirmed under a rigorous multi-seed protocol. It is also, on its own, an incomplete basis for an architecture recommendation: the ranking did not replicate under this paper's independent cross-condition protocol (Section 4.4), which also differs in spatial resolution. Thus the result does not isolate wall temperature or any other individual physical factor as the cause of the change. It does show that **a statistically well-powered single-trajectory comparison, however rigorously analyzed at the seed level, does not by itself establish**

a generalizable architectural property in this domain. Seed-level rigor and condition-level rigor are separate axes of evidence.

A plausible, though unproven, contributing mechanism is that spectral truncation's smoothing behavior and a local receptive field's edge sensitivity trade off differently depending on the specific interface geometry, thermal gradient magnitude, and vapor coverage extent present in a given trajectory — properties that vary meaningfully with wall superheat. We note this as a candidate explanation consistent with the observed pattern, not as an established causal mechanism; isolating it would require a dedicated study correlating per-trajectory physical statistics (interface curvature, vapor coverage fraction, thermal gradient magnitude) with per-trajectory architecture ranking, which the current two-condition independent test cannot support.

The exploratory same-trajectory 96×96 comparison (Section 4.7) produces directionally consistent results with the multi-trajectory cross-condition test (Section 4.4): both stored 96×96 configurations show positive T-FNO − U-Net paired differences on interface-temperature metrics, opposite to the tutorial-split 48×48 result. We treat this consistency as hypothesis-generating rather than confirmatory, because the 96×96 stored pipeline configurations differ from the 48×48 configuration in multiple factors beyond resolution (Section 3.8); no factorial design isolating resolution has been executed. The observation nevertheless strengthens the motivation for a future controlled study (Section 6).

5.2 The divergence penalty is the more robust of the two interventions tested

Unlike the interface/conservation ranking itself, the divergence penalty's conservation benefit transferred directionally across both the tutorial split and the independent conditions (Sections 4.3–4.4), even though only the tutorial-split result carries full statistical confirmation. This is consistent with the penalty targeting a specific, physically motivated quantity (velocity divergence) directly, rather than relying on an architectural inductive bias (locality) whose interaction with a given trajectory's specific physical conditions is less predictable. We view this as suggestive, not conclusive, evidence that explicit physical constraints generalize more reliably than architectural changes alone for this class of problem — a hypothesis for future, better-powered testing (Section 6). The differentiable Leray-projection approach of Li et al. (2026)[10] is a natural harder alternative: it constrains the velocity field to a divergence-free subspace, whereas the present work only penalizes divergence during training. This paper does not test that hard constraint, and it should not be interpreted as a comparison with it.

5.3 Physical output validity is necessary but architecturally orthogonal

The phase-bounding correction (Section 3.3) is, independent of any architecture ranking, a requirement for trustworthy autoregressive evaluation of any phase or volume-fraction field. The initial single-seed reversal it produced in rollout behavior (Section 4.2) is a cautionary example of exactly the single-condition/single-seed over-interpretation risk this paper's broader argument (Section 5.1) warns against, now demonstrated at the seed level rather than the trajectory level. We recommend both checks — physical-bound verification and multi-seed confirmation — as standard practice for future phase-resolved operator-learning evaluations.

5.4 CHF-motivated evaluation remains proxy-only

Across every trajectory available to this study, no sustained ground-truth dry-area event was observed (Section 4.5). The dry-area proxy therefore functions in this paper as an autoregressive-stability stress test and a demonstration of the physical-bounding pitfall (Section 4.2), not as an evaluation of CHF detectability, which would require at minimum one trajectory with a verified stable-to-film-boiling transition and, ideally, a synchronized wall-heat-flux measurement independent of the model's own output.

5.5 Limitations

Dimension	Limitation	Status in this work
Statistical power, cross-condition	Independent test uses 2 trajectories, 5 seeds; minimum exact p=.0625, no Holm-significant cross-condition result	Directionally consistent, explicitly reported as unconfirmed
Physical-condition sampling	2 held-out conditions cannot characterize population-level regime variation	Explicitly scoped; not generalized beyond these conditions

Dimension	Limitation	Status in this work
Resolution	96×96 cross-condition training is converged; 384×384 is feasibility-only (1 epoch)	Native-resolution convergence is future work
Phase representation	Binary `dfun>0` mask, not continuous volume fraction; omits surface tension/contact-angle dynamics	Acknowledged; continuous-target ablation is future work
CHF validity	No trajectory contains a verified stable-to-CHF transition or synchronized heat-flux series	Proxy-only throughout; no detection claim made
Divergence-penalty selection	$\lambda_{\text{div}} = .30$ is the upper boundary of the tested range	Section 6 future work extends the range
Conservation scope	Only velocity divergence is penalized; energy/momentum conservation are not explicitly constrained	Noted as an open direction
Architecture coverage	No comparison to transformer-based operators (e.g., Bubbleformer) or hard-constrained conservation layers	Explicitly out of scope this cycle; noted as related work
Hardware	Tutorial-split results use Apple M2/MPS; cross-condition results use NVIDIA T4/CUDA	Both reported explicitly per experiment; not cross-validated on identical hardware

Dimension	Limitation	Status in this work
Reproducibility	Retained tutorial and intervention data/result artifacts are checksummed; complete cross-condition per-seed histories and checkpoints are unavailable	Stored-result self-test is reproducible; full raw-data-to-checkpoint reproduction remains unavailable
Exploratory resolution comparison	The 48×48 and 96×96 stored configurations differ in temporal frame range, hardware, source revision, and checkpoint availability in addition to spatial resolution; no factorial resolution-only control has been executed	Reported as descriptive only; a fully controlled resolution experiment is identified as future work (Section 6)

5.6 Practical and methodological implications

For practitioners, the safest actionable guidance from this study is: do not select an architecture for phase-resolved boiling forecasting based on a single trajectory's results, regardless of how many seeds or how rigorous the statistical test at that trajectory. Where an explicit conservation requirement exists, a targeted physical penalty (Section 4.3–4.4) currently appears to be the more robust lever than architectural locality alone, though this too awaits confirmation at higher cross-condition statistical power. Where interface fidelity is paramount and only a single, representative operating condition is relevant (e.g., a fixed-design cooling system with a known, narrow operating envelope), the tutorial-split result may still be locally informative — but should not be assumed to generalize to a different operating point without direct testing, exactly as this paper's own result demonstrates.

For the broader field, we argue the methodological contribution — predeclared decision gates, physical-validity checks, and cross-condition replication — is at least as valuable as any specific architecture finding reported here, and we encourage its adoption as a standard evaluation practice for neural operator claims on chaotic, multiphysics, condition-varying systems.

6. Future Work (Explicitly Non-Blocking for This Submission)

The following are identified, well-motivated next steps that would strengthen this line of work but are not required to support the claims made above, each of which is scoped and evidenced independently of them:

- Extending the λ_{div} sensitivity sweep beyond 0.30 to identify whether an interior optimum exists or whether performance continues to improve, plateaus, or degrades at higher values.
- Expanding the cross-condition test beyond two held-out trajectories toward the full ten-trajectory archive, to move from a directionally-consistent-but-underpowered result toward a properly powered, Holm-corrected cross-condition significance test.
- Training the cross-condition comparison to full convergence at native 384×384 resolution, beyond the current one-epoch feasibility check.
- A dedicated mechanism study correlating per-trajectory physical statistics (interface curvature, vapor coverage, thermal gradient magnitude) with per-trajectory architecture ranking, to move Section 5.1's candidate explanation from plausible to tested.
- Comparison against transformer-based operators (e.g., Bubbleformer-style architectures) and hard-constrained/projection-based conservation layers, as a stronger baseline set than the soft-penalty approach used here.
- A continuous (non-binary) phase-fraction target, to test whether the interface/conservation trade-off pattern is specific to the signed-distance-derived binary mask used in this study.
- Acquisition of, or synthetic construction of, at least one trajectory with a verified stable-to-CHF transition and synchronized heat-flux labeling, to move the CHF-motivated proxy toward an actual validated detection evaluation.
- A factorial resolution experiment — training the same model configurations on the same trajectories with matched temporal windows, matched hardware, matched checkpointing, and a predeclared architecture×resolution statistical interaction test — to determine whether the directional differences observed in Section 4.7 are attributable to spatial resolution specifically, or to the other pipeline factors that currently co-vary with it.

A concrete, ready-to-execute prompt for the first two items (the highest-leverage, most directly comparable extensions of this paper's existing protocol) is provided as a companion document.

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7. Reproducibility and Artifact Manifest

All data sources are checksummed (Section 3.1), and the locally retained tutorial and intervention experiments include their configurations, seeds, and training histories. The cross-condition experiment is represented locally by its compact audited summary rather than complete cloud-side per-seed histories and checkpoints; this boundary is documented in the artifact manifest. All statistical comparisons use a single, unchanged, audited implementation (Section 3.6) across every reported result. The release package includes per-seed JSON results where retained, benchmark outputs, training curves, and acquisition/preprocessing scripts for both data formats. The prior public `v1.0.4` repository release is archived at Zenodo, DOI `https://doi.org/10.5281/zenodo.21858198`. That archive is a version-specific snapshot and does not remove the stated raw-data, checkpoint, cross-condition-provenance, or field-snapshot limitations.

The exploratory same-trajectory 96×96 comparison (Section 4.7) is archived on the `experiment/resolution-control-n11` branch at commit `eb7884e`. This archive includes per-seed `config.yaml`, `results.json`, and training curve files for all 11 T-FNO and 11 U-Net seeds, the merged `benchmark_results.json` and `benchmark_summary.csv`, and the `resolution_control_analysis.json` containing the computed paired differences and Holm-adjusted p-values. The 96×96 model checkpoints are not committed (Kaggle-resident only). The control matrix documenting all pipeline differences is retained in `audit_n11/control_matrix.md`. These stored artifacts are reproducible at the analysis and summary level from the committed JSON files; full end-to-end reproducibility from raw data requires externally hosted data, checkpoints, and the Kaggle GPU environment.

Bibliographic verification status

Thirteen identifiable works were independently checked against primary publication records and are listed below. Additional author-intended references cannot be added without their source records; no bibliographic details were inferred. See `BIBLIOGRAPHY_VERIFICATION.md` for the evidence boundary.

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